

Lesson Content: Coordinates in Scratch 3

1. Introduction to Coordinates

Coordinates are pairs of numbers used to locate a point on a plane or surface. In a two-dimensional coordinate system, such as the Cartesian coordinate system, each point is represented by an ordered pair (x, y) . The x-coordinate represents the horizontal position of the point, while the y-coordinate represents the vertical position. Together, these coordinates pinpoint the exact location of the point on the plane.

In Scratch 3, every sprite moves on a **Stage** using a **coordinate system**. This system helps Scratch know **where** to place a sprite and **how** it should move.

The coordinate system has **two directions**:

- **X-axis** → moves the sprite **left and right**
- **Y-axis** → moves the sprite **up and down**

Each position on the Stage is written as **(x, y)** .

2. The Scratch Stage and Centre Point

- The Scratch Stage is divided into four parts.
- The **centre of the Stage** is at **$(0, 0)$** .
- When a sprite is at the centre, its x value is 0 and its y value is 0.
- The full coordinate range in Scratch:
 - X ranges from **-240 to 240**
 - Y ranges from **-180 to 180**
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Examples:

- $(0, 0)$ → centre of the Stage
- $(100, 0)$ → right side
- $(-100, 0)$ → left side
- $(0, 100)$ → top
- $(0, -100)$ → bottom

3. Understanding X and Y Values

- **Positive x values** move the sprite to the **right**
- **Negative x values** move the sprite to the **left**
- **Positive y values** move the sprite **up**
- **Negative y values** move the sprite **down**

Learners can see the **x and y values** of a sprite under the Stage while dragging the sprite with the mouse.

4. Motion Blocks Used with Coordinates

Scratch provides special motion blocks that use coordinates:

- **go to x: y:**
Moves the sprite instantly to a specific position.
- **change x by ()**
Moves the sprite left or right by a number.
- **change y by ()**
Moves the sprite up or down by a number.
- **set x to () / set y to ()**
Places the sprite on a fixed x or y position.

5. Demonstration Activity (Teacher-led)

1. Add a sprite to the Stage.
2. Drag the sprite and observe the changing x and y values.
3. Use the block **goes to x: 0 y: 0** to return the sprite to the centre.
4. Change the x value to move the sprite left and right.
5. Change the y value to move the sprite up and down.

6. Learner Practice Activity

Learners try the following tasks:

- Move the sprite to **(100, 50)**
- Move the sprite to **(-120, -80)**
- Make the sprite move right by 20 using **change x by**
- Make the sprite move up by 30 using **change y by**

7. Real-Life Connection

Coordinates are used in:

- **Maps and GPS**
- **Video games**
- **Computer graphics and animations**

Just like maps use directions, Scratch uses coordinates to control movement.

8. Guided Practice Project: Moving a Ball Using Coordinates in Scratch 3

Time: 30 minutes

Project Title: *Coordinate Ball Challenge*

Step 1: Set Up the Project (5 minutes)

1. Open **Scratch 3**.
2. Click **Choose a Sprite** → select the **Ball** sprite.
3. Click **Choose a Backdrop** → select the **Xy-grid** backdrop.
4. Ensure learners can see the **x and y values** below the Stage.

The grid helps us see where the sprite is positioned.

Step 2: Position the Ball Using Coordinates (5 minutes)

1. Click on the **Ball** sprite.
2. Go to the **Motion** blocks.
3. Drag the block **go to x: () y: ()**.
4. Add this script:

```
when green flag clicked  
go to x: 0 y: 0
```

5. Change the values and observe how the ball moves:
 - (100, 50)
 - (-100, -50)

x moves left/right and **y** moves up/down.

Step 3: Move the Ball Using Glide Blocks (8 minutes)

1. From **Motion**, drag the **glide 1 secs to x: () y: ()** block.
2. Connect it below the previous block:

```
when green flag clicked  
go to x: 0 y: 0  
glide 1 secs to x: 120 y: 80  
glide 1 secs to x: -120 y: 80  
glide 1 secs to x: -120 y: -80  
glide 1 secs to x: 120 y: -80
```

3. Click the green flag and watch the ball move smoothly.

glide makes movement smooth compared to “go to”.

Step 4: Add Obstacles (7 minutes)

1. Click **Choose a Sprite** → select a **Box**, **Wall**, or **Rock** sprite.
2. Rename it **Obstacle**.
3. Place the obstacle at any position on the Stage.
4. Duplicate the obstacle to add more challenges.

Obstacles make the game **more interesting and challenging**.

Step 5: Add Penalty Conditions (5 minutes)

1. Click on the **Ball sprite**.
2. Go to **Control** and **Sensing** blocks.
3. Add this script:

```
forever
if <touching Obstacle?> then
go to x: 0 y: 0
```

4. Test the project by letting the ball touch the obstacle.

This is a **condition** and the ball is “penalized” by returning to the start.

Extension (Optional)

- Add a **sound** when the ball touches an obstacle.
- Add a **score** that reduces by 1 when the obstacle is touched.
- Change glides speed to make the game harder.

Reflection questions:

- What happens when x is positive?
- What happens when y is negative?
- What block moves the sprite smoothly?

Expected Learning Outcome

By the end of this guided practice, learners will:

- ✓ Use **go to x: y:** to position sprites
- ✓ Use **glide** blocks for smooth movement
- ✓ Understand **coordinates** in Scratch
- ✓ Apply **conditions** using obstacles

Summary

- Scratch uses an **x–y coordinate system**
- The centre of the Stage is **(0, 0)**
- X controls **left and right**
- Y controls **up and down**
- Coordinates help sprites move accurately